AR-Mini · Human in the loop

One maker: Teddy. Eight AI agents propose, check and plan. Teddy builds, measures and decides. · Sofia · 07.10 → 01.11.2026

AI team · 8 agents, each with its own context and files

Agents never touch the hardware and never mark anything as tested. Every output goes to Teddy first.

Coordinator
Claude

- reads the loop tracker each morning
- today: 3 priorities
- Teddy's task list for the day
- cuts scope if a release slips
- tracks R1 / R2 / R3 dates
- flags blocked loops

→ daily plan

Mechanical
Claude · ChatGPT

- OpenSCAD changes from v1.9
- small test sections first
- clearance + collision checks
- print orientation, settings
- hydrostatics: draft and trim
- what to measure after printing

→ .scad + .3mf + hypothesis

Electrical
Claude · ChatGPT

- pin → pin wiring tables
- fuse tree before first power
- what to measure + expected value
- breadboard sheets
- current budget per branch
- datasheet checks

→ wiring + checklist

Software
Claude · CURSOR

- ESP32-S3 firmware, PlatformIO
- bench test sketches
- CSV log over serial, 10 Hz
- heat-seek → approach → lift
- serial commands for tests
- commit message for Teddy

→ code to flash

Logistic
Claude

- BOM statuses, orders
- delivery times to Sofia
- what is missing for the next test
- print queue
- filament stock
- shop alternatives: fast vs. cheap

→ BOM + shopping list

Media
Claude · Gemini

- shot list for every test
- iPhone settings, framing
- video script for the portfolio
- file names + captions
- B-roll list for the story
- edit plan for 30–60 s clips

→ shot list

Test & Data
Claude · Gemini

- test protocol + pass criterion
- CSV → charts
- hypothesis vs. result
- reviews test video
- repeat count, tolerance
- what the fail means

→ verdict to Teddy

Portfolio
Claude

- FAILS_AND_FIXES + LOG
- MIT Maker Portfolio text
- only what has a photo/video
- marks human vs. AI work
- README + decisions
- final 01.11 package

→ portfolio draft

The loop · every arrow passes through Teddy

Teddy · AI agent

Hypothesis — Teddy: what we expect, with a number
→ **Proposal** — agent: CAD, wiring, code, order
→ **Review** — Teddy: approve, change or reject
→ **Build** — Teddy: print, assemble, wire, flash
→ **Measure** — Teddy: test + iPhone photo/video
→ **Analyse** — agent: data vs. hypothesis
→ **Decide** — Teddy: pass/fail, close the loop

next version · logged in the AR-Mini Loop tracker · goal: hours per loop, not days

TEDDY · human in the loop

Builds, measures and decides. The only one who touches the robot, and the only one who can close a loop.

1 3D printer care
- dry and load PETG / PLA
- clean the bed, check the first layer
- nozzle and belt checks, print log
- empty the bin, check spool path
- photo of first layer + part

with Mechanical · Logistic

2 Assembly
- heat-set inserts, M3 / M4 / M5
- hull sections + neutral silicone
- beams, arm, cradle, power box
- torque by feel, threadlocker
- fit check against CAD

with Mechanical

3 Breadboard prototype
- DevKit-Lipo + BTS7960 + servo
- wire from the breadboard sheet
- bench_v1: pot → motor, servo
- PWM_MAX 150 on first runs
- label every wire

with Electrical · Software

4 Soldering
- headers, terminals, XT60
- wires + heat-shrink
- fume extractor on, third hand
- pull test every joint
- continuity after soldering

with Electrical

5 Electrical measurements
- continuity, no shorts BEFORE power
- voltages at the fuse block
- current draw, battery voltage
- fuse rating vs. real current
- photo of every reading

with Electrical · Test & Data

6 Dimensions + weight
- calipers on every part
- scale: parts and full robot (4.7 kg)
- numbers into the tracker
- hole spacing on bought parts
- compare to CAD, note delta

with Mechanical

7 Water tests
- leak test: dry paper inside
- bathtub: float, draft, tilt
- pool: swim, then lift Pesho
- electronics dry-run first
- towel, dry box, timer

with Test & Data

8 Firmware on the board
- flash over USB-C
- serial CSV log to a file
- set servo IDs (Waveshare)
- keep a known-good build
- note version in the tracker

with Software

9 iPhone photos
- every part next to a ruler
- every fail, before / after the fix
- the meter display for every number
- clean background, side light
- AirDrop to the laptop daily

with Media

10 iPhone video
- 4K 30 fps, landscape
- 5 s before and after the action
- top-down rig over the table / pool
- slo-mo for water and impacts
- one clip per loop

with Media

11 Lab + safety
- parcels: check against the BOM
- LiPo: charge in the bag, supervised
- smoke or heat: power off, then photo
- clean bench at night
- printer runs overnight, Teddy sleeps

with Logistic

12 Decide + record
- review every AI proposal
- close the loop in the tracker
- commit to GitHub, real dates
- next hypothesis
- what did this loop teach?

with Coordinator · Portfolio

Tools

subscriptions: Claude Max · Cursor Premium · ChatGPT Plus · Gemini Plus

Claude · CURSOR · ChatGPT · Gemini

capture

iPhone

Rules

- AI proposes, a measurement decides.
- Portfolio numbers come only from the tracker.
- Unknown size or pin = a measuring task, never a guess.
- Releases: R1 13.10 · R2 22.10 · R3 29.10 · submit 01.11.